

LAITH ASSAF

East Lansing, MI | (248) 480-5783 | assaflai@msu.edu | [linkedin.com/in/laith-assaf-](https://www.linkedin.com/in/laith-assaf-) | github.com/Novapool

EDUCATION

Michigan State University, College of Engineering

BS Computer Science, AI/ML | Dean's List x3

East Lansing, MI

August 2022 - May 2027

- **Coursework:** Data Structures & Algorithms, Databases & Cloud Computing, Software Design

EXPERIENCE

PC Technician Intern

Onsite Support Team, Delta Dental

May 2026 - Present

Okemos, MI

- Deployed and configured **150+** workstations across a new corporate office buildout, imaging endpoints via **Ivanti** MDM; building an automated ticket creation workflow in **Power Automate** to streamline IT operations

IT and Media Services Support Associate

MSU College of Osteopathic Medicine

May 2023 - Present

East Lansing, MI

- Supported 100+ classroom AV and conferencing systems across 4 buildings serving **1,200+ students**, resolving 50-70 tickets weekly and maintaining **99.5% uptime** through a documented diagnostic methodology

PROJECTS

Pokemon Battle Agent - Reinforcement Learning | *PyTorch, PPO, MCTS*

May 2026 – Present

- Trained a competitive battle agent through **behavior cloning** on human replay data, **anchored PPO** fine-tuning of an MLP policy/value network over structured per-Pokemon token observations, and **determinized MCTS** at decision time, reaching a **93% win rate** against scripted baselines
- Built a parallel simulation and evaluation harness on the **Pokemon Showdown** engine, running **747 rated ladder games** across **11 milestones** against a pre-registered evaluation methodology
- Ran a code review that uncovered **four confounds** invalidating prior results (greedy decoding explained the reported search gain; concessions inflated ladder win rate **~7pp**), then restated every affected result

Agentic Development System | *Claude Code, subagents, hooks, MCP*

January 2026 – Present

- Designed a **10-agent orchestration pipeline** (planner → parallel builders → reviewer, tester, simplifier, docs auditor) using a single transient plan file as the handoff contract, applied across **44 repositories**
- Instrumented **11 lifecycle hook events** to enforce conventions automatically, and authored a tiered documentation protocol requiring every milestone to declare a **pre-registered, testable success gate**

SoundSense - Real-Time Audio Classification | *YAMNet, Flask* | **2nd Place, MHacks**

October 2025

- Placed **2nd of 300+ participants at MHacks 2025** building a real-time audio classifier on **YAMNet** CNN and MemryX edge accelerators, hitting **95% accuracy** at <50ms latency across 521 AudioSet classes

AI-Assisted Comparative Market Analysis | *scikit-learn, pandas*

June 2026 – Present

- Built a sold-price model over metro Detroit **MLS** history reaching **13.3% MAPE**, then diagnosed the ceiling as missing features (square footage on only **17%** of records) rather than model capacity

LEADERSHIP

Michigan State University | AI Club

Workshop Coordinator & Project Lead

East Lansing, MI

January 2025 - Present

- Delivered 8+ technical workshops on **local LLM deployment** (Ollama, LM Studio), agent orchestration, and **PyTorch** fundamentals to 500+ attendees, and led a 4-student team shipping an **LLM-powered** course performance predictor

TECHNICAL SKILLS

AI/ML: PyTorch, scikit-learn, pandas, NumPy, PPO / reinforcement learning, MCTS, behavior cloning, CNNs

AI Engineering: LLM APIs (Anthropic, OpenAI), agent orchestration, MCP, local inference (Ollama), evaluation

Languages & Backend: Python, C++, C#, TypeScript, SQL, FastAPI, Flask, Express, Node.js, REST APIs

Data & Infrastructure: PostgreSQL, Firebase, Streamlit, Docker, Linux, Git, Railway, Vercel